Julian Karlbauer M.Sc.

Systems-focused Research Engineer
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Specializing in Human-Computer-Interaction, Machine Learning, and Extended Reality (XR) systems. Expert in developing and deploying research prototypes into real-world applications, evidenced by work at Siemens, Audi, and Silicon Valley startups. Experienced HCI researcher with publications in leading conferences and journals.

EXPERIENCE

Sabbatical — Latin America, Middle East, Europe, Asia
Travelling and Volunteering — 06/2025 – present
A selection of my experiences includes: (i) contributing to journalism in the Middle East resulting in two published newspaper articles, (ii) extensive endurance and alpine trekking, (iii) living in a Buddhist Monastery, (iv) farm work.

Phygtl Inc. — Palo Alto, USA *(part-time, remote)*
Founding Engineer — 12/2024 – 06/2025
Equity-based founding role. Worked on a multiplayer social AR platform enabling GenAI based 3D co-creation. Responsible for frontend implementation, software architecture, code quality, and Android/iOS releases. Collaborated directly with the executive team to define the technical roadmap.

Self Employed — Ulm, Germany *(part-time, remote)*
Independent Consultant for XR Systems — 11/2024 – 04/2025
Built a decentralized emergency simulation prototype targeted at Apple Vision Pro for an industrial client. The prototype was demoed at the GISEC Global 2025 exhibition.

Ulm University — Ulm, Germany
Postgraduate HCI Researcher — 01/2024 – 10/2024
Worked as a Research Associate under Prof. Dr. Enrico Rukzio fully funded by the German Research Foundation (DFG). I Conducted research on Generative-AI supported 3D authoring in AR to enhance digital manufacturing workflows. I demoed my work at ACM SCF24. I co-authored a paper at the ACM MUM24 conference.

Siemens AG — Munich, Germany
Student Researcher — 04/2023 – 11/2023
Built an impactful AR prototype that caused traction in the research group and was recognized as a future candidate to be demoed to Roland Busch (CEO), a demo at Formnext (leading convention on Additive Manufacturing), and development into a commercial Siemens product. The system is an important step towards usable Digital Twin for industrial metal 3D printing. It enables anomaly detection, 3D visualization, data labelling, and data-lake integration to enhance User Experience in industrial scale Additive Manufacturing

Audi AG — Ingolstadt, Germany
Software Engineering Intern — 03/2022 – 09/2022
Worked in an engineering team to develop a mobile cross-platform AR application designed for Audi internal communication. My focus was on Unity3D based iOS development and interaction design.

TriCAT GmbH — Ulm, Germany
R&D Intern — 03/2021 – 06/2021
Worked independently on an AR system on HoloLens 2 for in-situ real-time object recognition and 3D localization using CNNs and environment meshes with a compute efficient remote GPU architecture to tackle unsolved customer challenges. Through my work I pioneered the use of AI tools in the company.

Ulm University Department of Media Informatics — Ulm, Germany
Research Assistant — 04/2018 – 03/2021
Co-authored papers at CHI [2] and ToCHI [3]. Developed VR Research Prototypes, conducted User Studies, and tutored students.

EDUCATION

Ulm University — Ulm, Germany
M. Sc. in Media Informatics — 10/2020 – 03/2024
Thesis Title: *Show, Don't Tell: Enhancing Quality Assurance in Additive Manufacturing through Augmented Reality (Grade 1.0)*

Norwegian University of Science and Technology — Trondheim, Norway
Visiting Student in Generative AI — 08/2021 – 12/2021
Thesis Title: *Raw Audio Music Emotion Composition using Mood Annotated Spotify Playlists (Grade 1.0)*

Ulm University — Ulm, Germany
B. Sc. in Media Informatics — 04/2016 – 07/2020
Thesis Title: *An Exploration of Foveated Blue Light Filters in Virtual Reality (Grade 1.0)*

ACADEMIC SERVICES

- Conference Reviewer CHI 2025
- Served as a student volunteer at the ACM SCF 2024 conference.

PERSONAL ACHIEVEMENTS

- Completed over 1,000km of solo high-altitude and long-distance trekking, including 500km of the Camino de Santiago (21 days) and self-supported navigation of the Peruvian Andes (> 5,000m altitude).
- Volunteered on a private farm in Thailand. My tasks included picking and processing of coffee, general property maintenance, and electronics repairs.
- Spent 2 weeks in a silent Theravada Buddhist Monastery practicing extensive daily meditation, property maintenance, and participating in traditional monastic life.
- Accompanied a journalist in the Middle East resulting in two published newspaper articles. I was responsible for capturing footage, assisting in interviews, and mediating interviewing partners.
- Solo performances of self-composed pieces on classical guitar as part of classical concerts.

ACADEMIC PUBLICATIONS

[1] Tobias Drey, Nico Rixen, **Julian Karlbauer**, and Enrico Rukzio. 2024. VRCreatIn: Taking In-Situ Pen and Tablet Interaction Beyond Ideation to 3D Modeling Lighting and Texturing. In Proceedings of the International **Conference on Mobile and Ubiquitous Multimedia (MUM '24)**. Association for Computing Machinery, New York, NY, USA, 24–35. https://doi.org/10.1145/3701571.3701580

[2] Teresa Hirzle, Fabian Fischbach, **Julian Karlbauer**, Pascal Jansen, Jan Gugenheimer, Enrico Rukzio, and Andreas Bulling. 2022. Understanding, Addressing, and Analysing Digital Eye Strain in Virtual Reality Head-Mounted Displays. **ACM Trans. Comput.-Hum. Interact.** 29, 4, Article 33 (August 2022), 80 pages. https://doi.org/10.1145/3492802

[3] Tobias Drey, Jan Gugenheimer, **Julian Karlbauer**, Maximilian Milo, and Enrico Rukzio. 2020. VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction for 3D Sketching in Virtual Reality. In Proceedings of the 2020 **CHI Conference on Human Factors in Computing Systems (CHI '20)**. Association for Computing Machinery, New York, NY, USA, 1–14. https://doi.org/10.1145/3313831.3376628